# Terna Engineering College
## Computer Engineering Department

**Class: TE** **Sem.: VI**

**Course: System Security Lab**

# PART A

(PART A : TO BE REFFERED BY STUDENTS)

## Experiment No.06

**A.1 Aim:** Design a network and implement packet sniffing on telnet traffic using wireshark.

### A.2 Prerequisite:

1. Basic Knowledge of IP addresses, Port numbers, TCP and UDP Protocols.

### A.3 Outcome:

**After successful completion of this experiment students will be able to**

Apply basic network command to gather basic network information.

### A.4 Theory:

Wireshark, a network analysis tool formerly known as Ethereal, captures packets in real time and display them in human-readable format. Wireshark includes filters, color-coding and other features that let you dig deep into network traffic and inspect individual packets.

Features of Wireshark :

- Available for UNIX and Windows.
- Capture live packet data from a network interface.
- Open files containing packet data captured with tcpdump/WinDump, Wireshark, and a number of other packet capture programs.
- Import packets from text files containing hex dumps of packet data.
- Display packets with very detailed protocol information.

- Export some or all packets in a number of capture file formats.
- Filter packets on many criteria.
- Search for packets on many criteria.

| SNO. | NAME OF THE DEVICE | INTERFACE | IP ADDRESS | Subnet Mask | Default Gateway |
|---|---|---|---|---|---|
| 1. | Router 0 | g0/0 | 192.168.120.1 | 255.255.255.0 | -------- |
| 2. | PC | Fast Ethernet | 192.168.120.2 | 255.255.255.0 | 192.168.120.1 |

- Colorize packet display based on filters.
- Create various statistics.

**Capturing Packets**

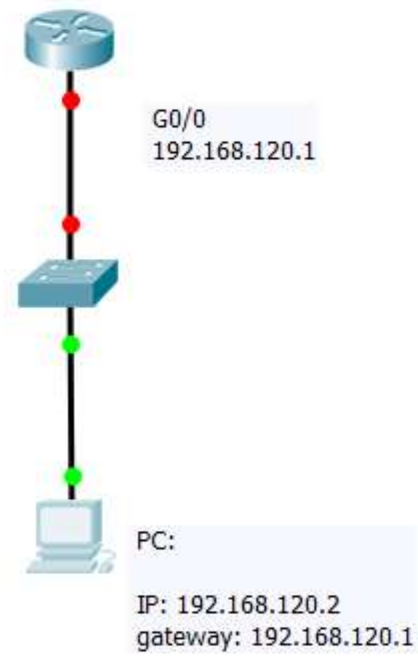
After downloading and installing wireshark, you can launch it and click the name of an interface under Interface List to start capturing packets on that interface. For example, if you want to capture traffic on the wireless network, click your wireless interface. You can configure advanced features by clicking Capture Options.

**Filtering Packets**

If you‘re trying to inspect something specific, such as the traffic a program sends when phoning home, it helps to close down all other applications using the network so you can narrow down the traffic. Still, you‘ll likely have a large amount of packets to sift through. That‘s where Wireshark‘s filters come in. The most basic way to apply a filter is by typing it into the filter box at the top of the window and clicking Apply (or pressing Enter). For example, type ―dns‖ and you‘ll see only DNS packets. When you start typing, Wireshark will help you autocomplete your filter.

**A5. Interface Configuration table**

**A6. Design**




# PART B

(PART B : TO BE COMPLETED BY STUDENTS)

***(Students must submit the soft copy as per following segments within two hours of the practical. The soft copy must be uploaded on the Blackboard or emailed to the concerned lab in charge faculties at the end of the practical in case the there is no Black board access available)***

| | |
|---|---|
| **Roll No.C54** | **Name: Quazi Md Moinuddin** |

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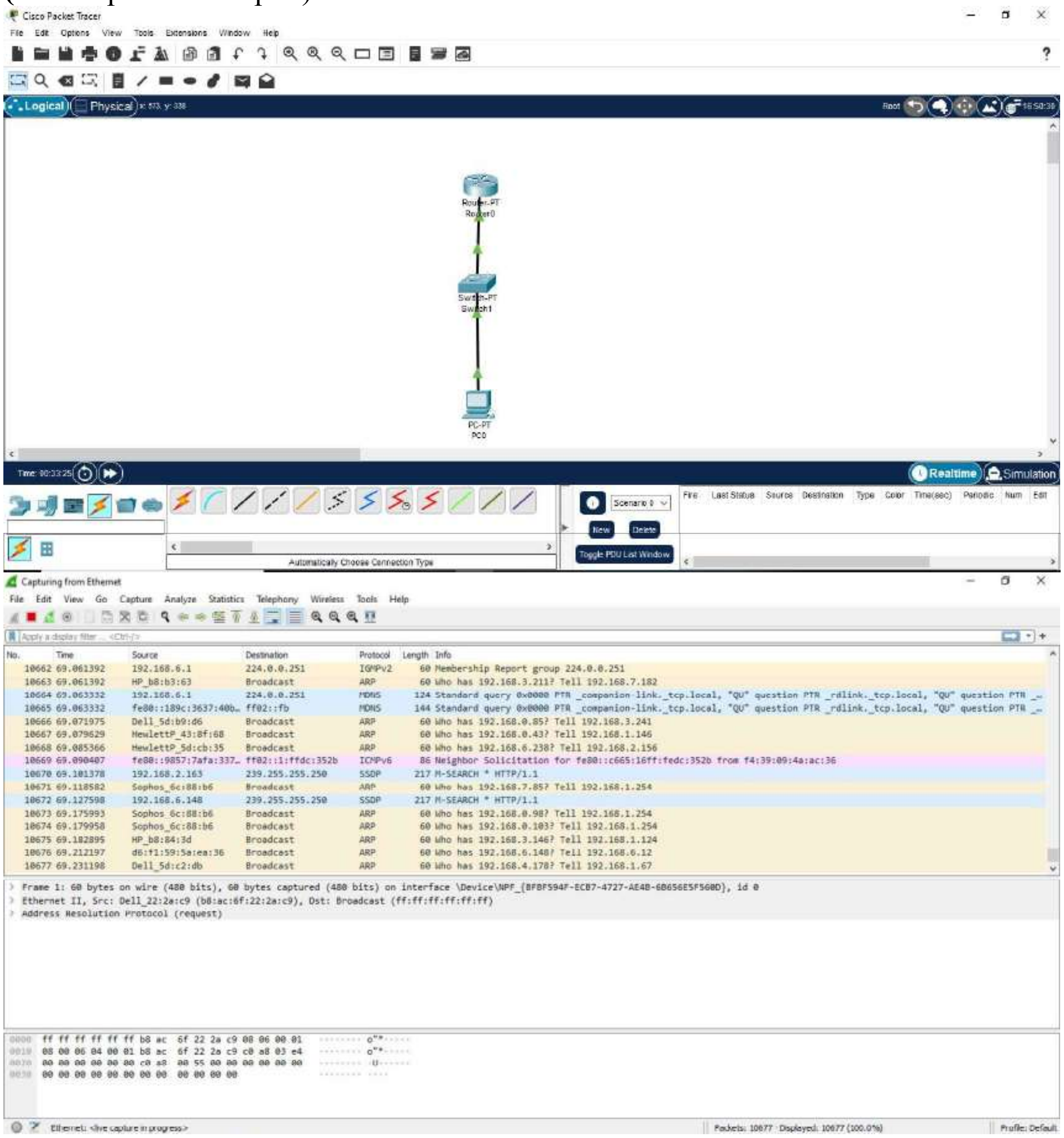
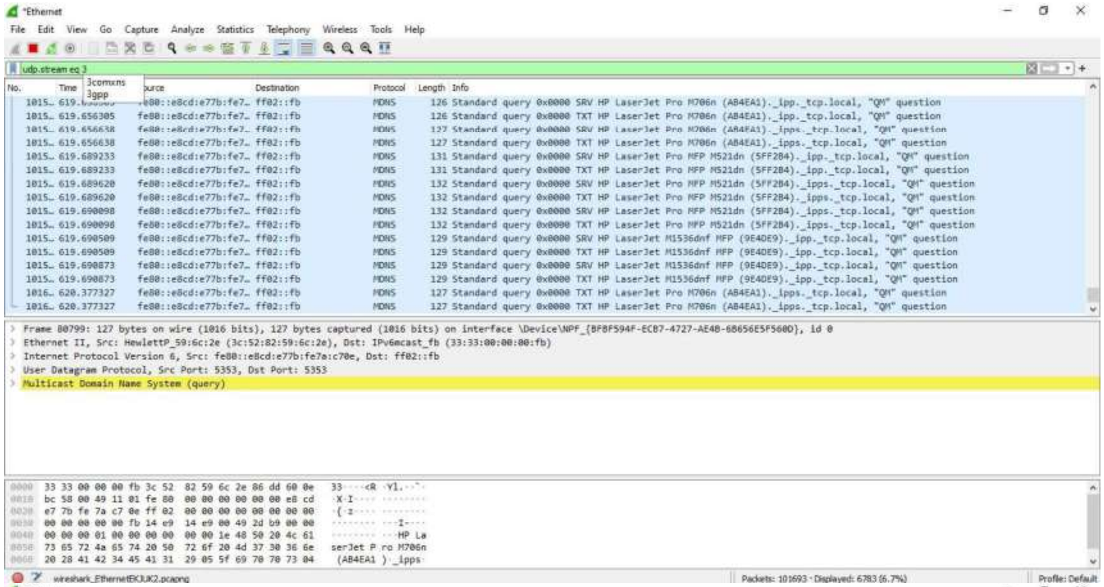
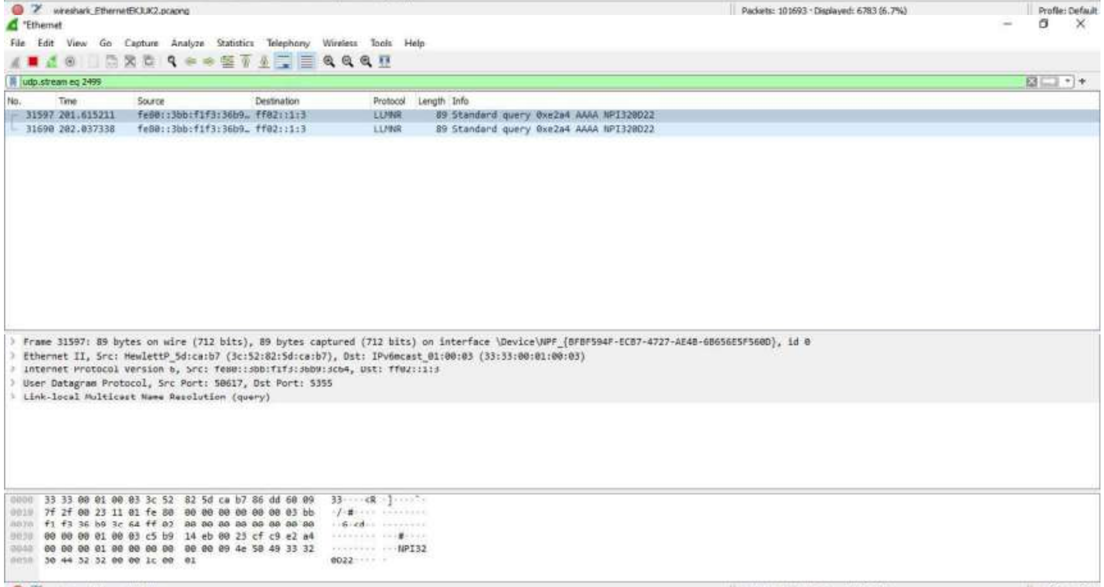
|---|---|
| **Class :TE-C** | **Batch :C3** |
| **Date of Experiment: 27-2-25** | **Date of Submission: 6-3-25** |
| **Grade :** | |

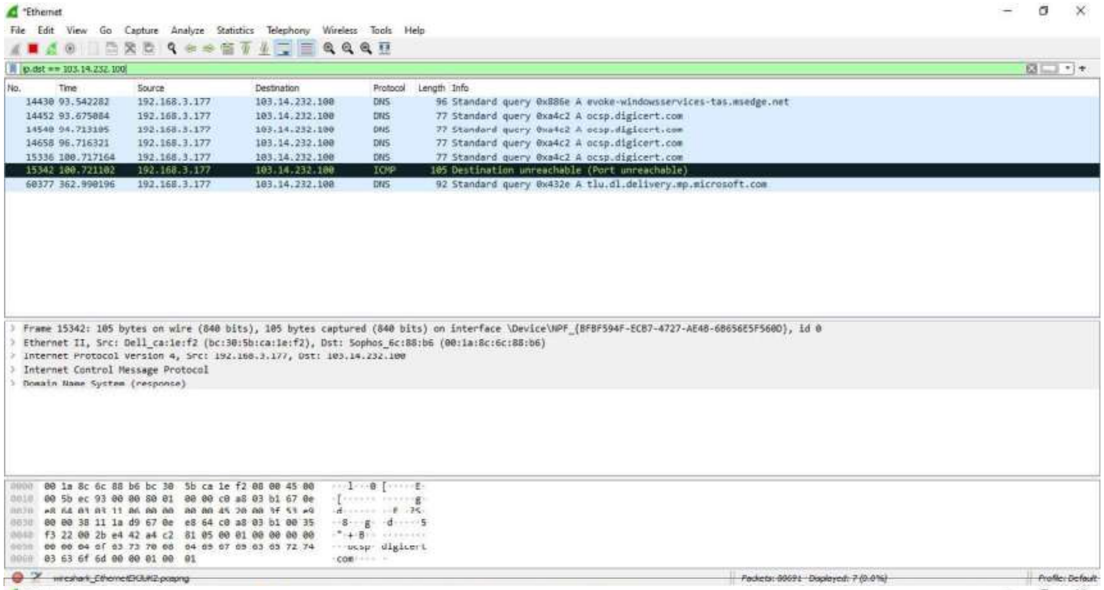
## B.1 Output

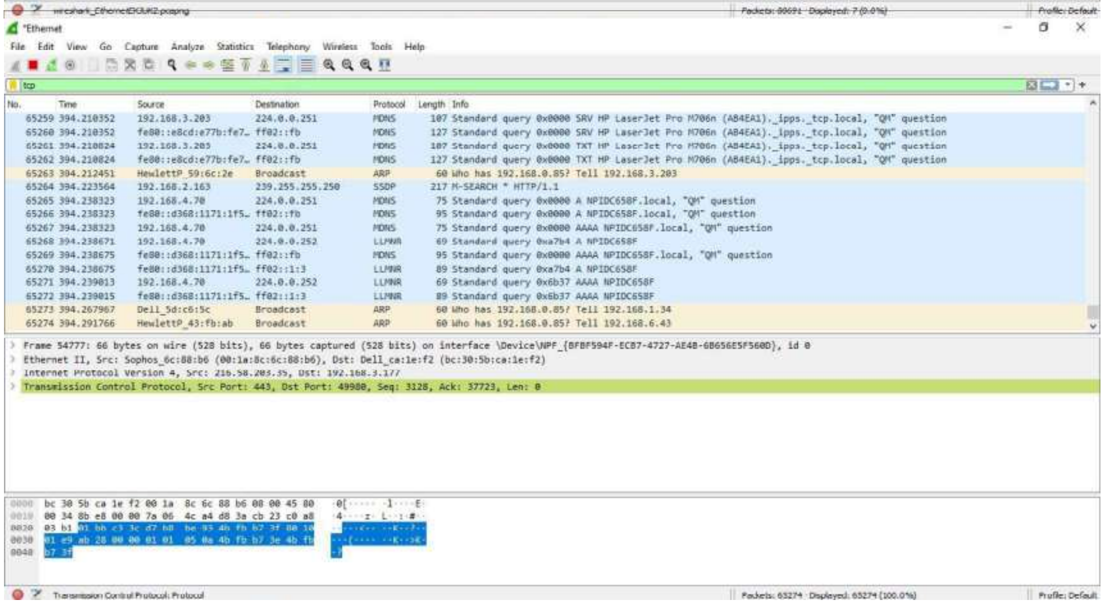
(add snapshot of output )

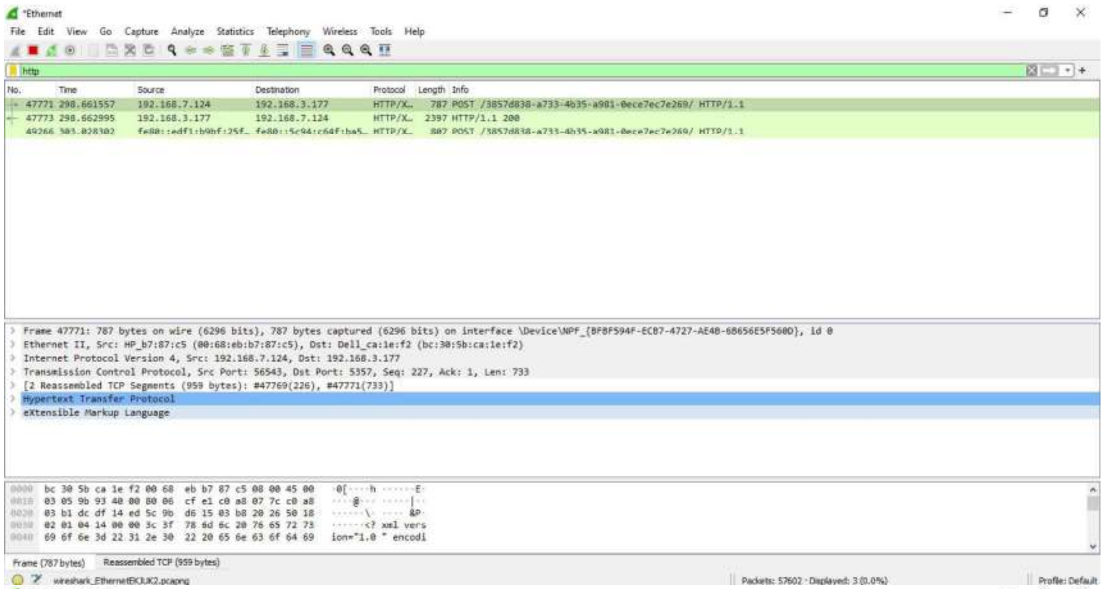


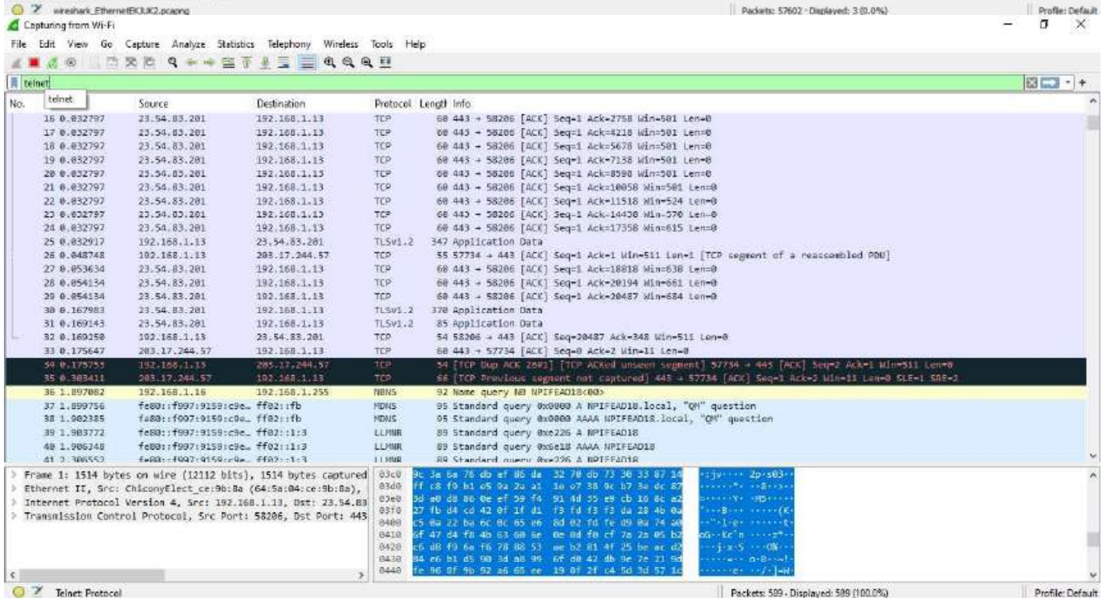

*Ethernet
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
udp.stream eq 3
No. Time Source Destination Protocol Length Info
Frame 80799: 127 bytes on wire (1016 bits), 127 bytes captured (1016 bits) on interface \Device\NPF_{BFBF594F-ECB7-4727-AE4B-6B656E5F560D}, id 0
Ethernet II, Src: HewlettP_59:6c:2e (3c:52:82:59:6c:2e), Dst: IPv6mcast_fb (33:33:00:00:00:fb)
Internet Protocol Version 6, Src: fe80::e8cd:e77b:fe7a:c70e, Dst: ff02::fb
User Datagram Protocol, Src Port: 5353, Dst Port: 5353
Multicast Domain Name System (query)
wireshark_EthernetEKJUK2.pcapng
Packets: 101693 · Displayed: 6783 (6.7%)
Profile: Default


*Ethernet
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
udp.stream eq 2499
No. Time Source Destination Protocol Length Info
31597 201.615211 fe80::3bb:f1f3:36b9… ff02::1:3 LLMNR 89 Standard query 0xe2a4 AAAA NPI320D22
31690 202.037338 fe80::3bb:f1f3:36b9… ff02::1:3 LLMNR 89 Standard query 0xe2a4 AAAA NPI320D22
Frame 31597: 89 bytes on wire (712 bits), 89 bytes captured (712 bits) on interface \Device\NPF_{BFBF594F-ECB7-4727-AE4B-6B656E5F560D}, id 0
Ethernet II, Src: HewlettP_5d:ca:b7 (3c:52:82:5d:ca:b7), Dst: IPv6mcast_01:00:03 (33:33:00:01:00:03)
User Datagram Protocol, Src Port: 50617, Dst Port: 5355
Link-local Multicast Name Resolution (query)


*Ethernet
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
ip.dst == 103.14.232.100
No. Time Source Destination Protocol Length Info
14430 93.542282 192.168.3.177 103.14.232.100 DNS 96 Standard query 0x886e A evoke-windowsservices-tas.msedge.net
14452 93.675084 192.168.3.177 103.14.232.100 DNS 77 Standard query 0xa4c2 A ocsp.digicert.com
14540 94.713105 192.168.3.177 103.14.232.100 DNS 77 Standard query 0xa4c2 A ocsp.digicert.com
14658 96.716321 192.168.3.177 103.14.232.100 DNS 77 Standard query 0xa4c2 A ocsp.digicert.com
15336 100.717164 192.168.3.177 103.14.232.100 DNS 77 Standard query 0xa4c2 A ocsp.digicert.com
15342 100.721102 192.168.3.177 103.14.232.100 ICMP 105 Destination unreachable (Port unreachable)
60377 362.990196 192.168.3.177 103.14.232.100 DNS 92 Standard query 0x432e A tlu.dl.delivery.mp.microsoft.com
Frame 15342: 105 bytes on wire (840 bits), 105 bytes captured (840 bits) on interface \Device\NPF_{BFBF594F-ECB7-4727-AE48-6B656E5F560D}, id 0
Ethernet II, Src: Dell_ca:1e:f2 (bc:30:5b:ca:1e:f2), Dst: Sophos_6c:88:b6 (00:1a:8c:6c:88:b6)
Internet Protocol Version 4, Src: 192.168.3.177, Dst: 103.14.232.100
Internet Control Message Protocol
Domain Name System (response)


*Ethernet
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
tcp
No. Time Source Destination Protocol Length Info
65259 394.210352 192.168.3.203 224.0.0.251 MDNS 107 Standard query 0x0000 SRV HP LaserJet Pro M706n (AB4EA1)._ipps._tcp.local, "QM" question
65263 394.212451 HewlettP_59:6c:2e Broadcast ARP 60 Who has 192.168.0.85? Tell 192.168.3.203
65264 394.223564 192.168.2.163 239.255.255.250 SSDP 217 M-SEARCH * HTTP/1.1
65265 394.238323 192.168.4.70 224.0.0.251 MDNS 75 Standard query 0x0000 A NPIDC658F.local, "QM" question
65273 394.267967 Dell_5d:c6:5c Broadcast ARP 60 Who has 192.168.0.85? Tell 192.168.1.34
65274 394.291766 HewlettP_43:fb:ab Broadcast ARP 60 Who has 192.168.0.85? Tell 192.168.6.43
Frame 54777: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface \Device\NPF_{BFBF594F-ECB7-4727-AE48-6B656E5F560D}, id 0
Ethernet II, Src: Sophos_6c:88:b6 (00:1a:8c:6c:88:b6), Dst: Dell_ca:1e:f2 (bc:30:5b:ca:1e:f2)
Internet Protocol Version 4, Src: 216.58.203.35, Dst: 192.168.3.177
Transmission Control Protocol, Src Port: 443, Dst Port: 49980, Seq: 3128, Ack: 37723, Len: 0
Packets: 65274 · Displayed: 65274 (100.0%)
Profile: Default




## B.2 Commands / tools used with syntax:

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Tools:

1. Wireshark: A packet sniffing tool used to capture and analyze network traffic.
2. Telnet Client: A Telnet client software installed on a computer to establish Telnet sessions with devices on the network.

Commands:

sudo wireshark

telnet telnet <hostname or IP address> telnet 192.168.1.100

## B.3 Question of Curiosity:

1. Which command is need to configure telnet in router?
2. What are the steps needed to extract data from sniffed traffic?
3. What type of packets to be filtered for accessing remote login username and password of a router?
4. What are the differences between Telnet and SSH in terms of security?
5. What filtering techniques can be used in Wireshark to analyze Telnet sessions?
6. How can encrypted protocols mitigate the risks identified in sniffed Telnet traffic?

CSS Expt 6

Q1] Which command is needed to configure Telnet router

→ To configure Telnet on a router, you need to enable the Vty (Virtual Teletype) lines & set a p for remote acces.

1. Enter global configuration mode:

Router > enable

Router # configure terminal

2. Acces the vty line configuration mode.

Router (config)# line vty 0 4

3. Set the password for telnet access.

Router (config-line)# password [your-password]

4. Enable login to use the password.

Router (config-line)# login

5. ~~Op~~ Exit the line config & return global

Router (config-line)# exit

7. Enable telnet & allow remote login acces.

Router (config)# transport input telnet

8. Exit configuration mode & Save changes.

Router (config)# exit

Router # write memory

2] What are the steps needed to extract data sniffed traffic?

→ 1. Capture the traffic.

- Use a packet sniffer like wireshark, tcpdump to capture network traffic on the target network interface

2. Follow traffic based protocol.

- If you're looking for specific data like HTTP, T filter the traffic protocol to narrow down captured packets. ~~eg.~~

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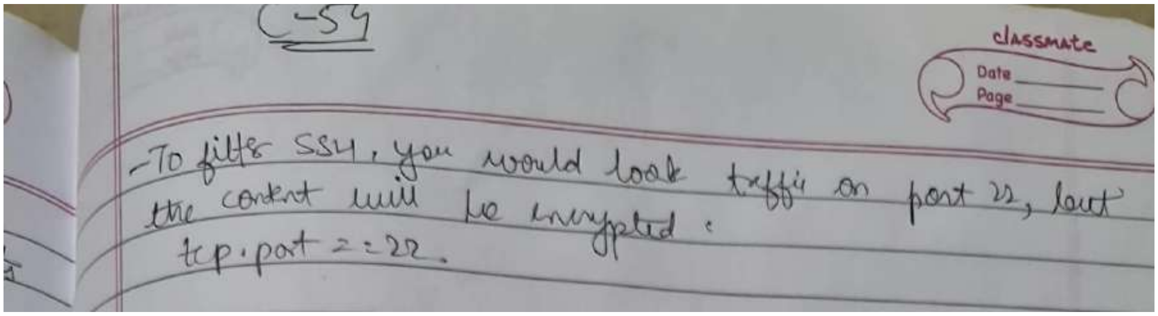
3. Inspect packet contents
- Once the traffic is captured, examine the packet contents to extract useful infoⁿ
- look for plaintext data or protocols that do not encrypt traffic.
4. Extract appⁿ data
- For telnet, you may find username, password if not encrypted
- For DNS, review DNS queries & response
- For FTP, you may extract username & passwords
5. Analyze & filter sensitive data
- Use additional tools like grep, awk or sed to automate extraction & parsing of the relevant data.

3) ⇒ 1. Telnet:
- Telnet sends data in plaintext (unencrypted), To Capture username, password during Telnet session, on port 23.
- Filter for TCP port 23 in your packet sniffer (eg wireshark
- look for login prompts & username/password pairs within the captured Telnet traffic.
tcp.port ==23.
2. FTP:-
- If the router is access via FTP which can als transmit credentials in text, filter for FTP traf on port 21.
- look for User & Pass commands in the FTP cont stream.
eg tcp.port ==21
3. SSH (secure shell):-
- SSH encrypts all traffic, including login credential As a result, sniffing SSH traffic for usernames & p is feasible without breaking the encryption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classmate
Date
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- To filter SSH, you would look traffic on port 22, but the content will be encrypted.
tcp.port == 22.

## B.4 Conclusion:

(Write appropriate conclusion.)

**In conclusion, designing a network and implementing packet sniffing on Telnet traffic using Wireshark requires careful planning, ethical considerations, and a focus on security. By analyzing captured packets and identifying potential vulnerabilities, organizations can improve their network security posture and mitigate risks associated with insecure protocols like Telnet.**